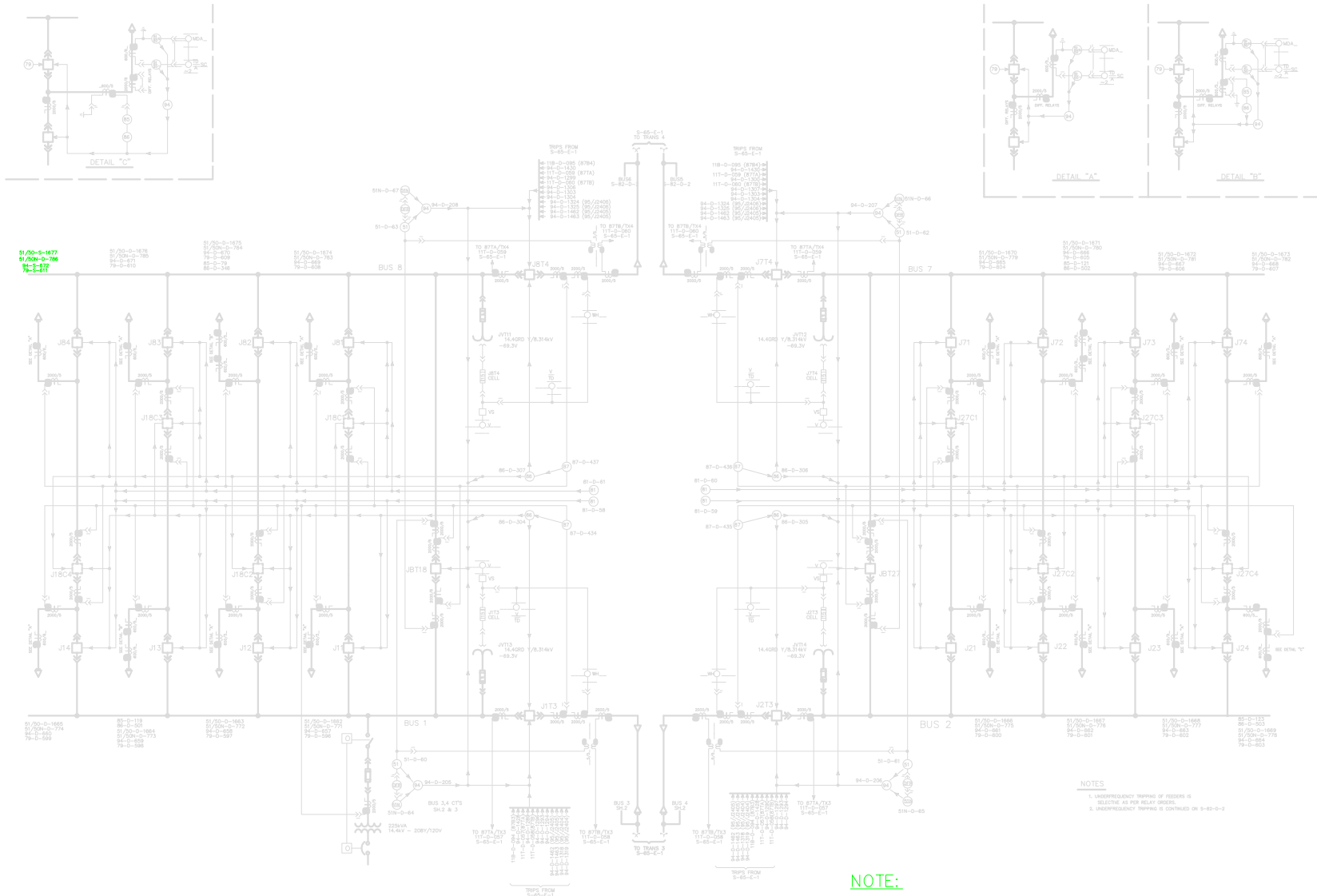




REV NUM	DATE	REVISION DESCRIPTION	DWN	DGN	CHK	ENG
A	12/15/25	ISSUED FOR RELAY REPLACEMENT	UV	UV	TM	TM

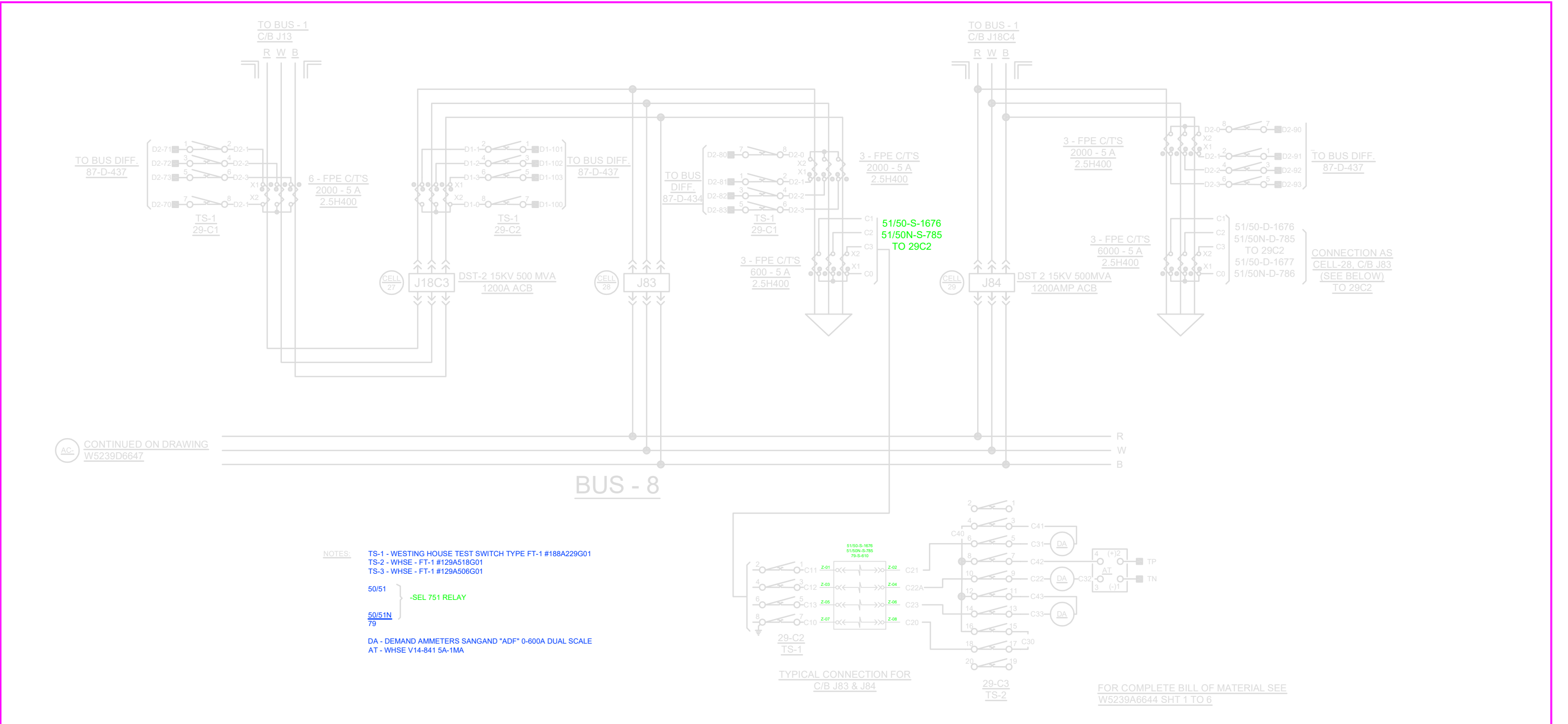
ENGINEER'S STAMP

SINGLE LINE DIAGRAM

DWN	DGN	CHK	ENG
UV	UV	TM	TM

DRAWING NUMBER:

S-82-D-1-REV1



CONTINUED ON DRAWING
W5239D6647

NOTES: TS-1 - WESTING HOUSE TEST SWITCH TYPE FT-1 #188A229G01
TS-2 - WHSE - FT-1 #129A518G01
TS-3 - WHSE - FT-1 #129A506G01
50/51 } -SEL 751 RELAY
50/51N }
79 }
DA - DEMAND AMMETERS SANGAND "ADF" 0-600A DUAL SCALE
AT - WHSE V14-841 5A-1MA

TYPICAL CONNECTION FOR
C/B J83 & J84

FOR COMPLETE BILL OF MATERIAL SEE
W5239A6644 SHT 1 TO 6

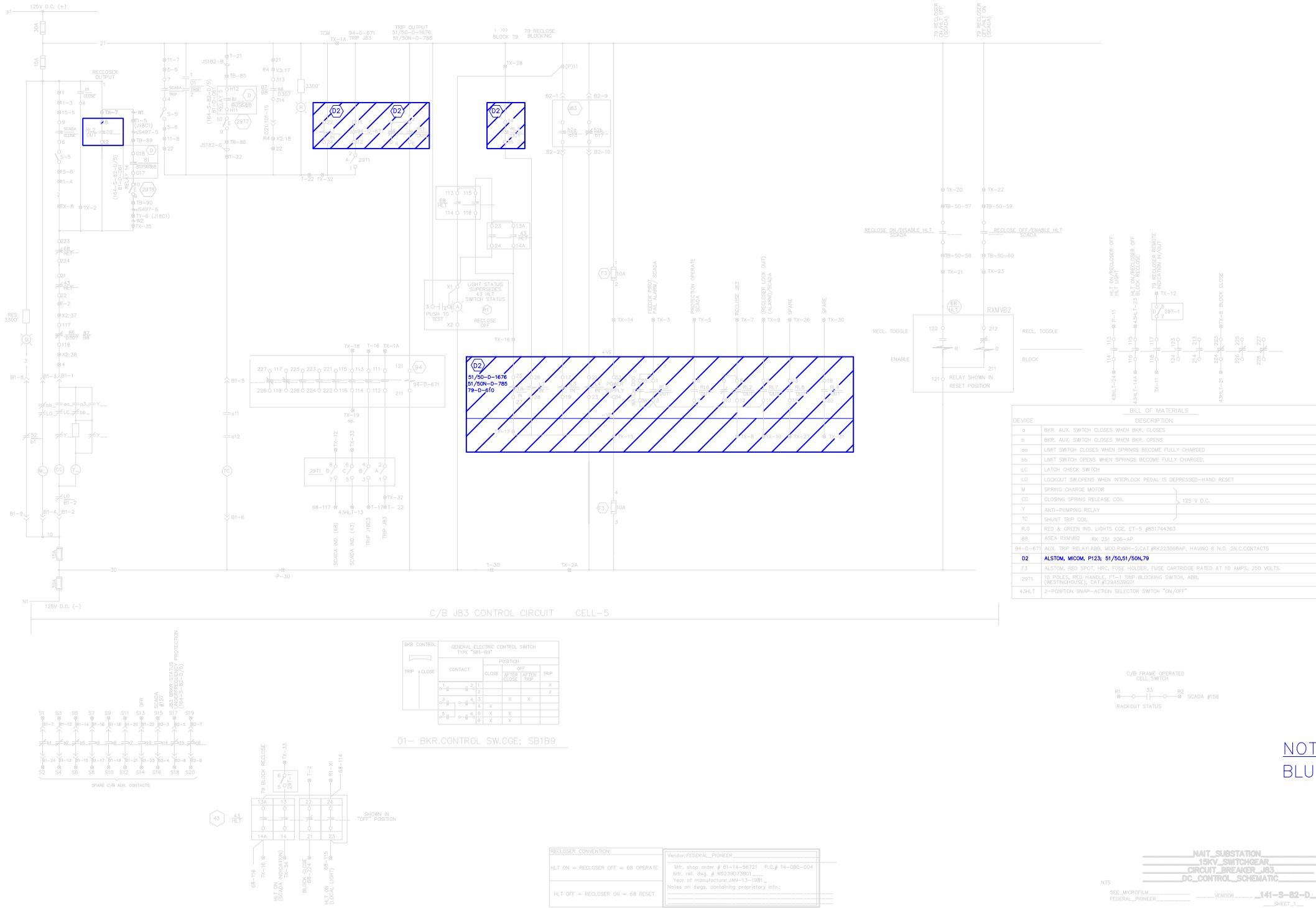
NOTE:
GREEN ELEMENT DENOTES CHANGE

NAIT SUBSTATION
15KV SWITCHGEAR
BUS 8: J83, J84
AC 3 LINE DIAGRAM

134-S-82-D
SHEET 1

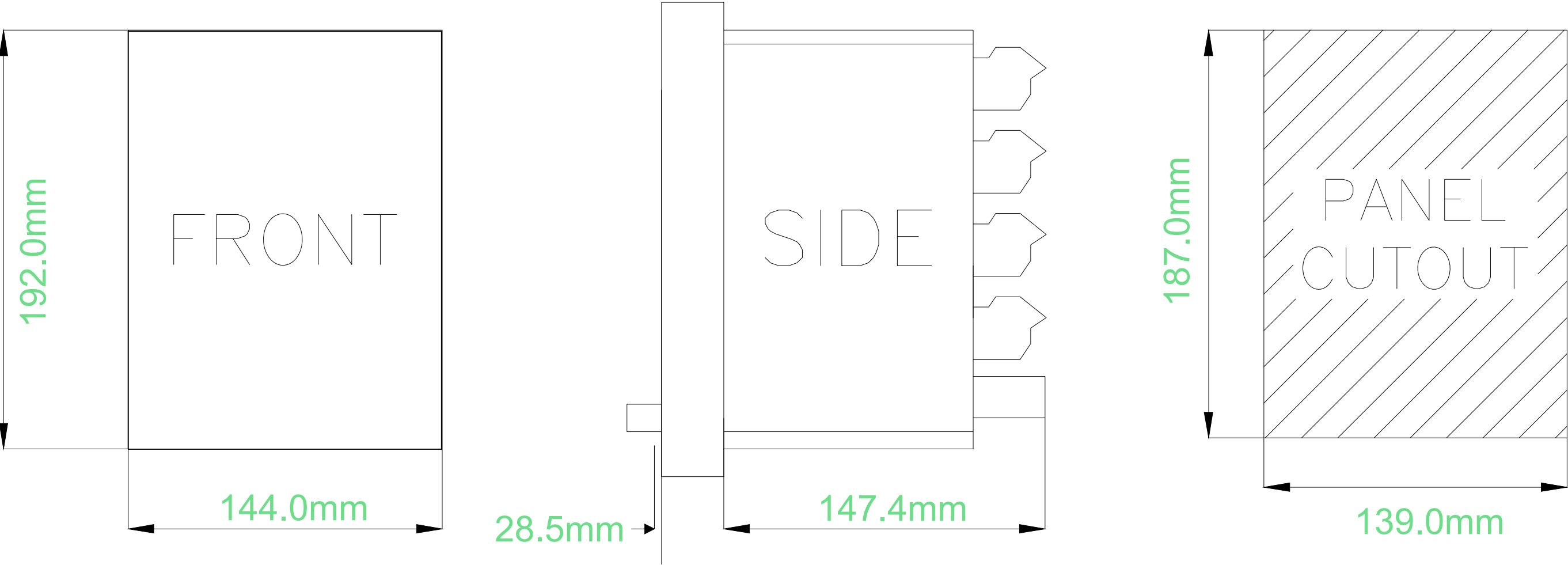
REV NUM	DATE	REVISION DESCRIPTION	DWN	DGN	CHK	ENG	REV NUM	DATE	REVISION DESCRIPTION	DWN	DGN	CHK	ENG	AC 3 LINE DIAGRAM			
A	12/15/25	ISSUED FOR RELAY REPLACEMENT	UV	UV	TM	TM								DWN	DGN	CHK	ENG
														UV	UV	TM	TM
														DRAWING NUMBER: 134-S-82-D-REV1			
														ENGINEER'S STAMP			

ENGINEER'S STAMP



REV NUM	DATE	REVISION DESCRIPTION	DWN	DGN	CHK	ENG	REV NUM	DATE	REVISION DESCRIPTION	DWN	DGN	CHK	ENG		ENGINEER'S STAMP	J-83 CONTROL SCHEMATIC -ADDITION			
1	12/15/25	ISSUED FOR RELAY REPLACEMENT	UV	UV	TM	TM													

CHASSIS



LEGEND

SCALE: 1:2

REV NUM	DATE	REVISION DESCRIPTION	DWN	DGN	CHK	ENG	REV NUM	DATE	REVISION DESCRIPTION	DWN	DGN	CHK	ENG		ENGINEER'S STAMP	SEL-751 LAYOUT			
0	12/15/25	ISSUED FOR RELAY REPLACEMENT	UV	UV	TM	TM										DWN	DGN	CHK	ENG
																UV	UV	TM	TM
																DRAWING NUMBER:			
																67-67-67-67			

Communications Card (Slot C)

NOTE: After any change, be sure to thoroughly test the settings.

Either the DeviceNet (see *Appendix H: DeviceNet Communications*) or the EIA-232/EIA-485 communications card is supported in Slot C. The EIA-232/EIA-485 card provides one serial port with one of the following two serial port interfaces:

- **Port 4A**, an isolated EIA-485 serial port interface
- **Port 4C**, a nonisolated EIA-232 serial port interface, supporting the +5 Vdc interface

Select either EIA-232 or EIA-485 functionality through use of the **Port 4 Setting COMM Interface**. *Table 2.3* shows the port number, interface, and type of connector for the two protocols.

Table 2.3 Communications Card Interfaces and Connectors

Port	Interface	Connectors
4A	EIA-485	5-pin Euro
4C	EIA-232	D-sub

I/O Card (4 DI/3 DO)

Supported in any nonbase unit slot (Slot C through Slot E), this card has four digital inputs, one Form B digital output (normally closed) and two Form C digital output contacts. *Table 2.12* shows the terminal designations.

Table 2.12 Four Digital Inputs/Three Digital Outputs (4 DI/3 DO) Card Terminal Designations

Side-Panel Connections Label	Terminal Number	Description ^a
01 OUT_01	01, 02	OUTx01, driven by OUTx01 SELOGIC control equation
02 OUT_02	03, 04, 05	OUTx02, driven by OUTx02 SELOGIC control equation
03 OUT_03	06, 07, 08	OUTx03, driven by OUTx03 SELOGIC control equation
04 IN_01	09, 10	INx01, drives INx01 element
05 IN_02	11, 12	INx02, drives INx02 element
06 IN_03	13, 14	INx03, drives INx03 element
07 IN_04	15, 16	INx04, drives INx04 element

^a x=3, 4, or 5 (e.g., OUT401, OUT402, etc. if the card is installed in Slot D).

I/O Card (3 DI/4 DO/1 AO)

NOTE: Analog output is self powered and has an isolated power supply.

NOTE: All digital inputs and digital outputs (including high-current, high-speed hybrid) connections are polarity neutral.

Supported in one nonbase unit slot (Slot C, D, or E), this card has three digital inputs, four digital outputs, and one analog output. Table 2.10 shows the terminal designations.

Table 2.10 I/O (3 DI/4 DO/1 AO) Card Terminal Designations

Side-Panel Connections Label	Terminal Number	Description ^a
01 } OUT_01	01, 02	OUTx01, driven by OUTx01 SELOGIC control equation
02 } OUT_02	03, 04	OUTx02, driven by OUTx02 SELOGIC control equation
03 } OUT_03	05, 06	OUTx03, driven by OUTx03 SELOGIC control equation
04 } OUT_04	07, 08	OUTx04, driven by OUTx04 SELOGIC control equation
05 } AO_01	09, 10	AOx01, Analog Output 1
06 } IN_01	11, 12	INx01, drives INx01 element
07 } IN_02	13, 14	INx02, drives INx02 element
08 } IN_03	15, 16	INx03, drives INx03 element
09 } INPUTS		

^a x=3, 4, or 5 (e.g., OUT401, OUT402, etc., if the card is installed in Slot D).

Power Supply Card PSIO/2 DI/3 DO (Slot A)

Select appropriate power supply option for the application:

- High Voltage: 110–250 Vdc, 110–240 Vac, 50/60 Hz
- Low Voltage: 24–48 Vdc

Select the appropriate digital input voltage option: 125 Vdc/Vac, 24 Vdc/Vac, 48 Vdc/Vac, 110 Vdc/Vac, 220 Vdc/Vac, or 250 Vdc/Vac.

This card is supported in Slot A of the SEL-751. It has two digital inputs and three digital outputs (two normally open Form A contact outputs and one Form C output). Table 2.1 shows the terminal designation for the PSIO/2 DI/3 DO card.

Table 2.1 Power Supply Inputs (PSIO/2 DI/3 DO) Card Terminal Designations

Side-Panel Connections Label	Terminal Number	Description
01 GND		Ground connection
02 +/H POWER	A01, A02	Power supply input terminals
03 -/N	A03, A04	OUT101, driven by OUT101 SELOGIC control equation
04 } OUT_01	A05, A06	OUT102, driven by OUT102 SELOGIC control equation
05 } OUT_02	A07, A08, A09	OUT103, driven by OUT103 SELOGIC control equation
06 } OUT_03	A10, A11	IN101, drives IN101 element
07 } IN_01	A12, A11	IN102, drives IN102 element
08 } IN_02		
09 } INPUTS		
A 100		

Current/Voltage Card (4 ACI/3 AVI)

⚠ WARNING

Before working on a CT circuit, first apply a short to the secondary winding of the CT.

MOT...x81x... (1 A phase, 1 A neutral CTs), or **...x85x...** (5 A phase, 5 A neutral CTs), or **...x82x...** (1 A phase, 5 A neutral CTs), or **...x86x...** (5 A phase, 1 A neutral CTs) or **...x83x...** (1 A phase, 200 mA neutral CTs), or **...x87x...** (5 A phase, 200 mA neutral CTs). Supported in Slot Z of the SEL-751, this card has current inputs for three-phase CTs, neutral current CTs, and 300 Vac voltage inputs for three-phase (wye or delta) PTs.

MOT...xL1x... (1 A phase, 1 A neutral CTs), or **...xL5x...** (5 A phase, 5 A neutral CTs), or **...xL2x...** (1 A phase, 5 A neutral CTs), or **...xL6x...** (5 A phase, 1 A neutral CTs) or **...xL3x...** (1 A phase, 200 mA neutral CTs), or **...xL7x...** (5 A phase, 200 mA neutral CTs). Supported in Slot Z of the SEL-751, this card has current inputs for three-phase CTs, neutral current CTs, and 8 Vac LEA voltage inputs for three-phase (wye or delta) PTs.

The 4 ACI/3 AVI current/voltage card inputs terminal designation for the card with LEA voltage inputs and the regular voltage inputs is as shown in Table 2.5.

2.8 Installation I/O Configuration

Table 2.5 Current/Voltage Inputs (4 ACI/3 AVI) Card Terminal Designations

Side-Panel Connections Label	Terminal Number	Description
	201	IA, A-Phase current input
	202	IB, B-Phase current input
	203	IC, C-Phase current input
	204	IN, Neutral current input
	205	VA, A-Phase voltage input
	206	VB, B-Phase voltage input
	207	VC, C-Phase voltage input
	208	N, common connection for VA, VB, VC